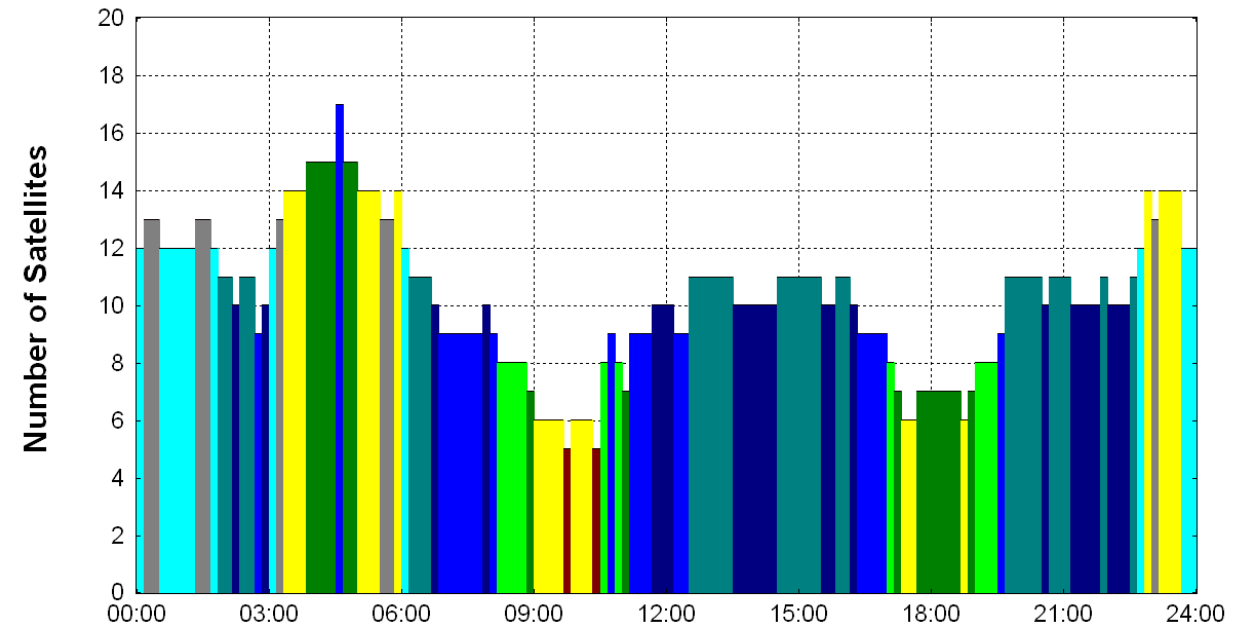


LAB 1: Planning of GNSS survey

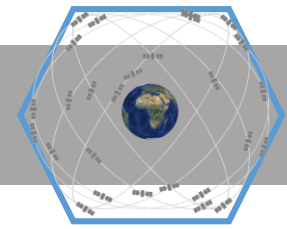
Visibility



Station North 45° 46' East 7° 28' Height 700m
Satellites 35 GPS 28 Glonass 7 [Almanac.alm]

Elevation cutoff 10° Obstacles 0%

Time 23/02/2007 00:00 - 24/02/2007 00:00 (GMT+2.0h)



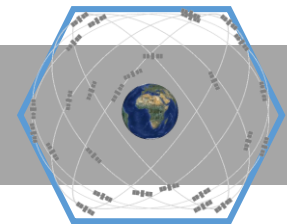
What are the most important actions?

- **on-the-field investigation**
- **analysis of satellite configuration**
- **to select the appropriate instruments**
- **materialization of points (in the field)**
- **to select the correct observation window**
- **to define the number and type of baselines (if necessary)**
- **starting survey sessions**

Green: actions which require a preliminary analysis

Red: actions which not require a preliminary analysis



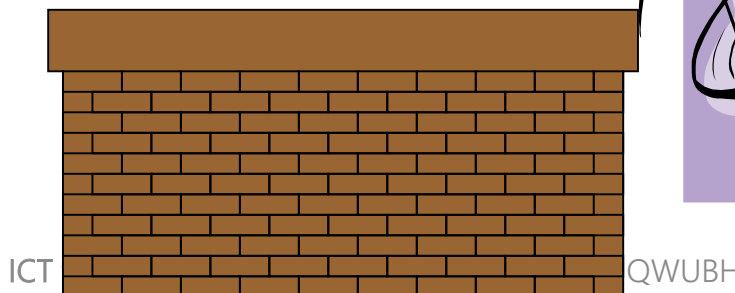
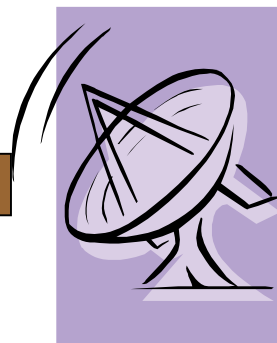


On-the-field investigation

The goal of the survey is to define the coordinates of one or more unknown points, creating a “visible” marker on the ground, for future activities.

Starting from the purpose of the survey, you have to consider:

- intervisibility of points
- no obstacles
- no electromagnetic interference
- stable structures



ICT

QWUBH

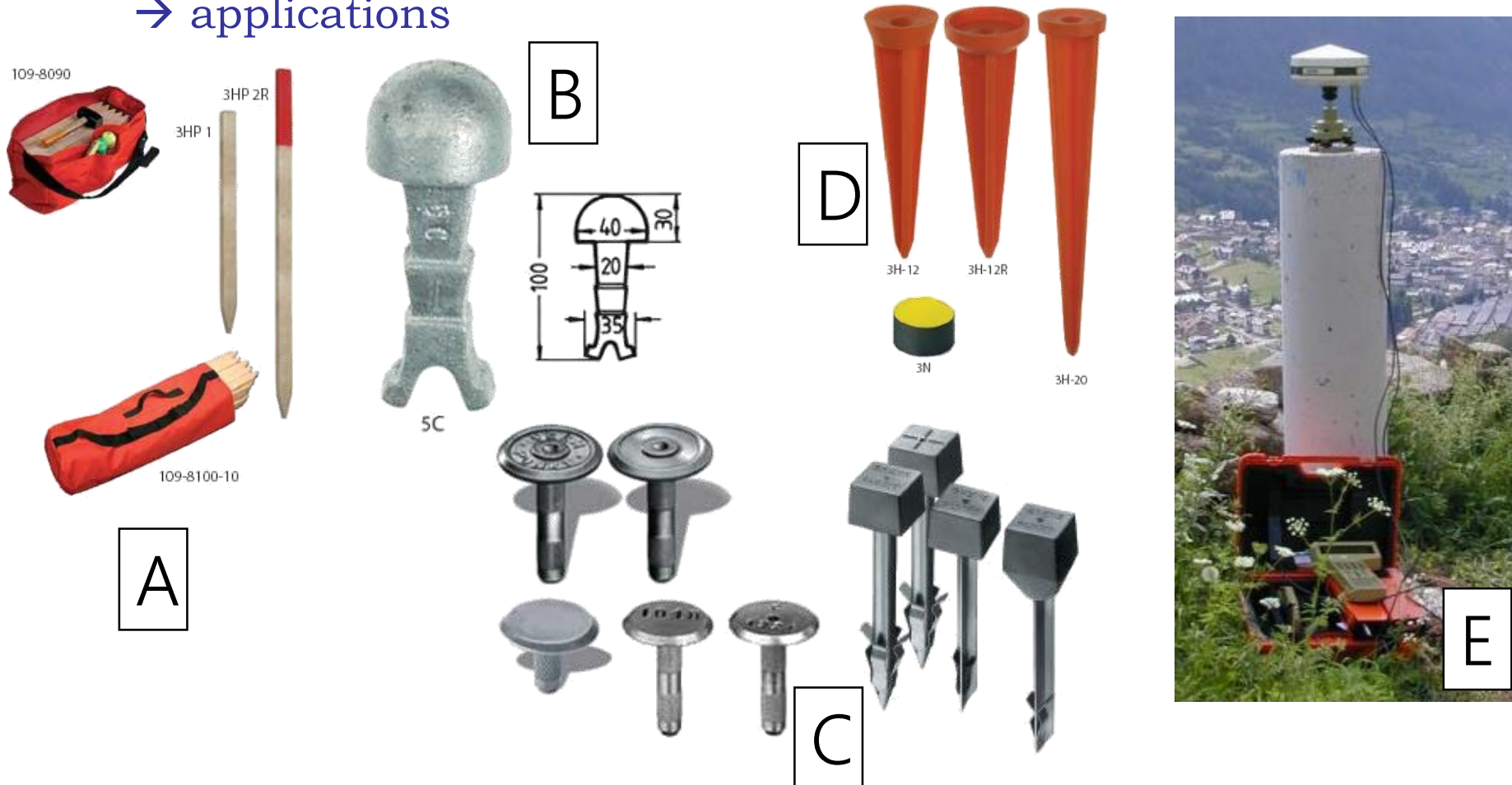
Prof. PIRAS

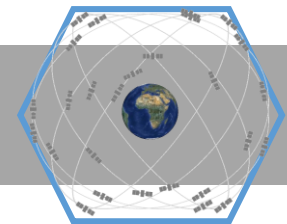


Type of supports for materialization

Selection of the correct support for the materialization

→ applications





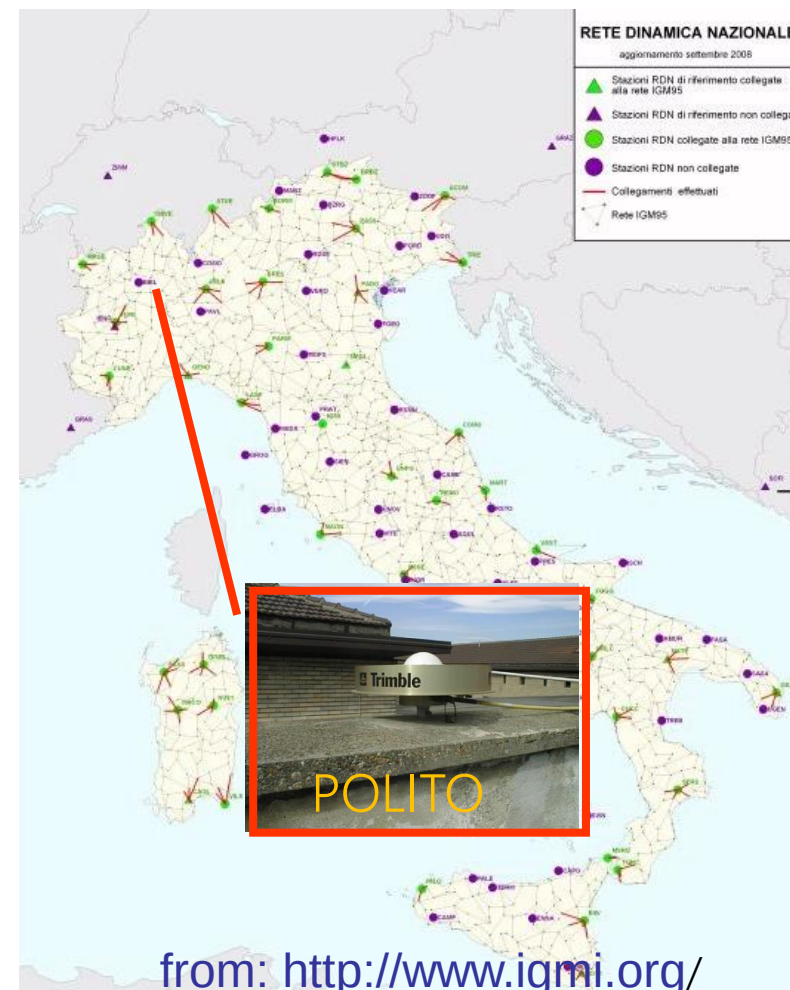
Type of supports for materialization

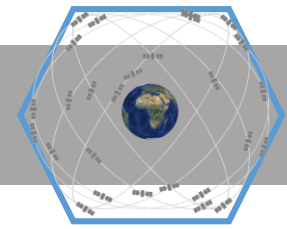
Rete dinamica nazionale IGM

(National Dynamic network)

- defines the R.F. ETRF2000 in a 'dynamic' way (in sense of continuous acquisition and recalculation)
- the Geoid is WGS84
- 99 CORS in Italy
- starting from:

1 January 2009



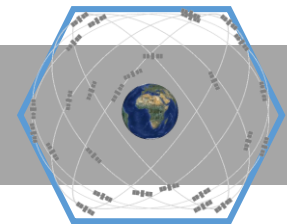


What is the planning?

In order to obtain a precise positioning, it is necessary to consider:

- obstacles
- reflective surfaces (e.g. multipath)
- presence of electromagnetic fields
- number of satellites
- elevation mask (cut off angle) (for tropo delay reduction)
- session length (5 min, 30 min, 1h, 5 h, 24 h)
- sampling rate (1", 30" ...), number of observation per second





What is the planning?

Navigation files

-YUMA

-SSF

-ALM

-Eph

-Rinex *.yyn

Station Editor

Station Name: **Site**

Position

Latitude: N 45 46

Longitude: E 7 28

Height: 700 [m]

Elevation Cutoff: 10 *

Time

Start Date: 23/02/2007 ..

Start Time: 00:00

Duration: 24 [h]

Interval: 10 [min]

Time Zone:

(GMT + 1.00 h) Amsterdam, Berlino, Berna, Roma, Stoccolma, Vienna

DST Difference GMT: 2.0 [h]

OK

Cancel

Apply

Delete

Obstacles...

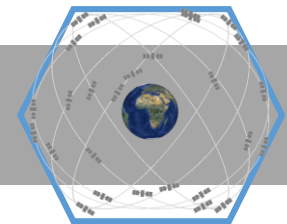
Map...

City...

Today

Time Zone...





Where are the almanac?

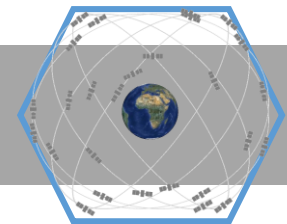
TRIMBLE:

<ftp://ftp.trimble.com/pub/eph/almanac.alm>

IGS:

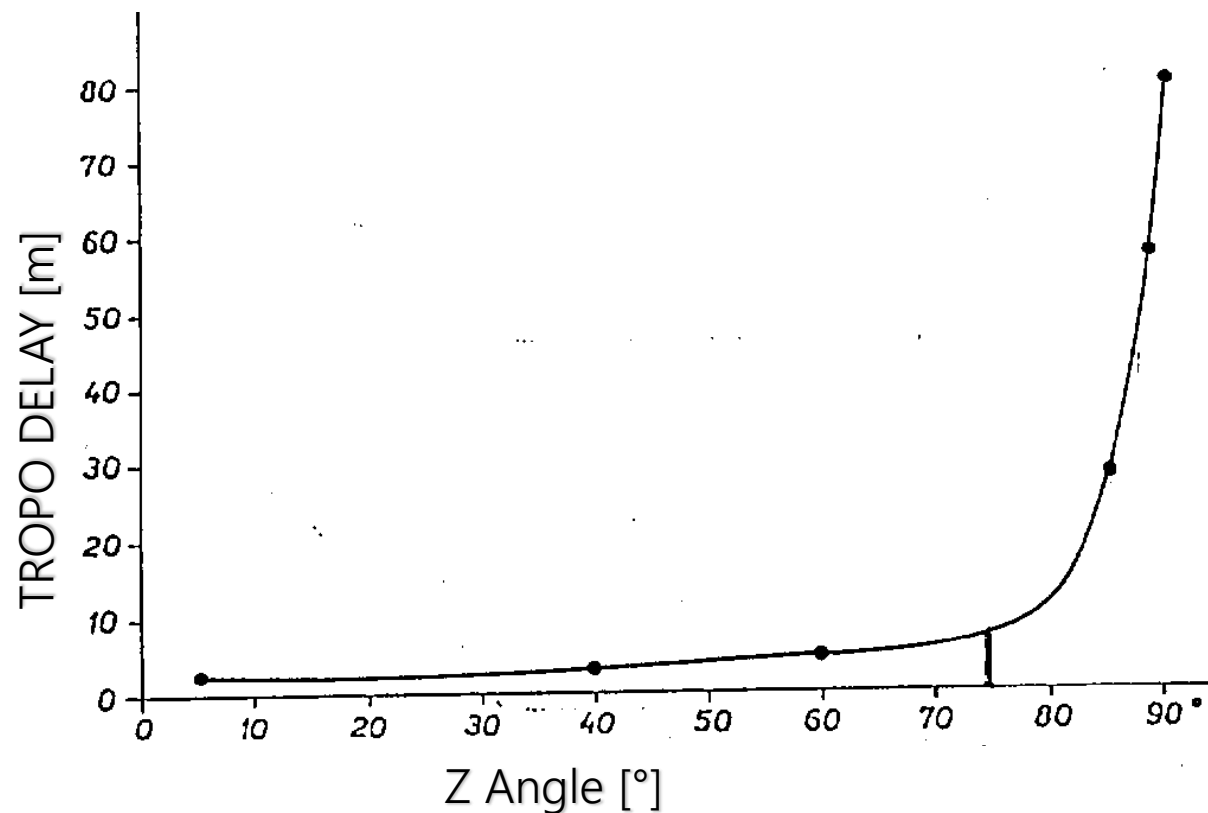
<ftp://igs.ensg.ign.fr/pub/igs/products/>

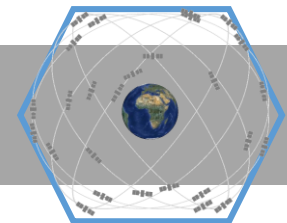




How to obtain a good positioning?

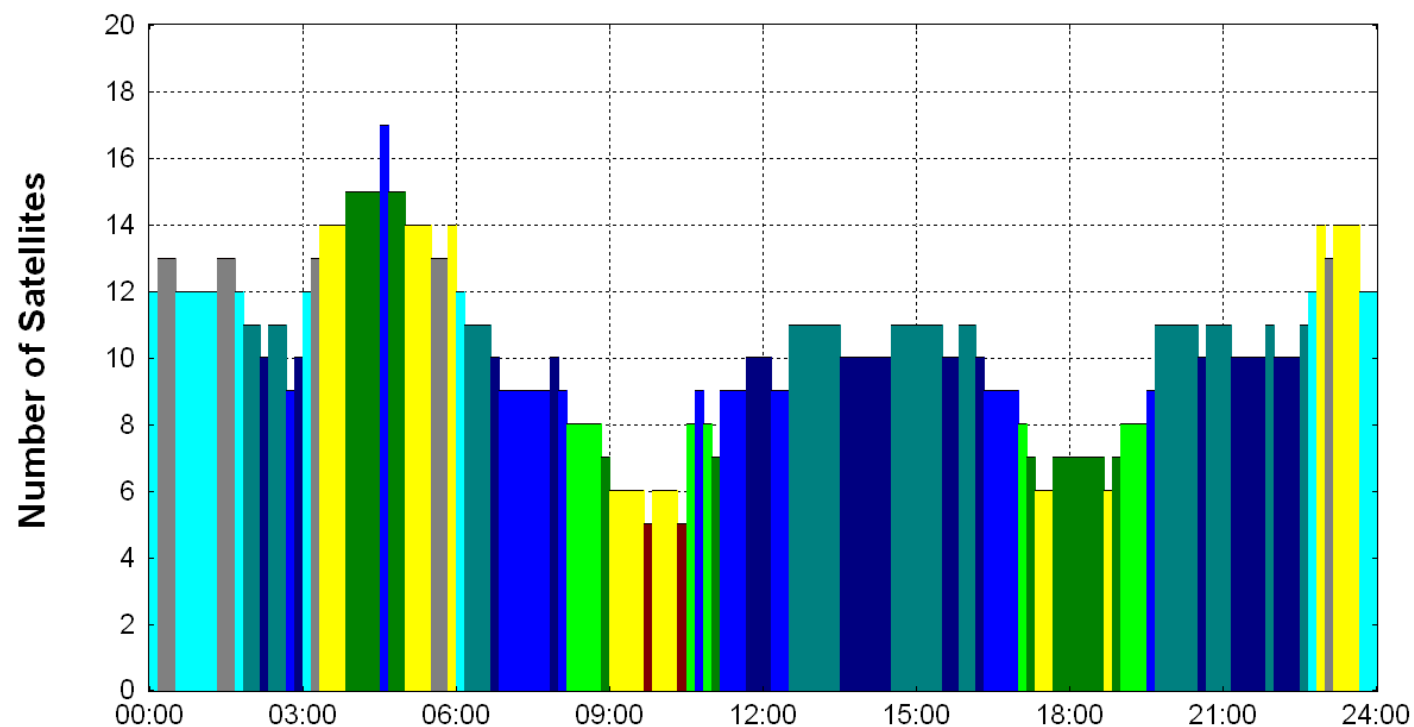
- ELEVATION MASK (or CUT OFF ANGLE)





Visibility plot

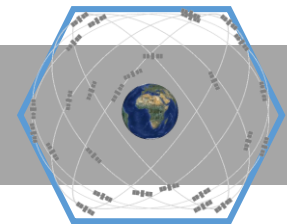
Visibility



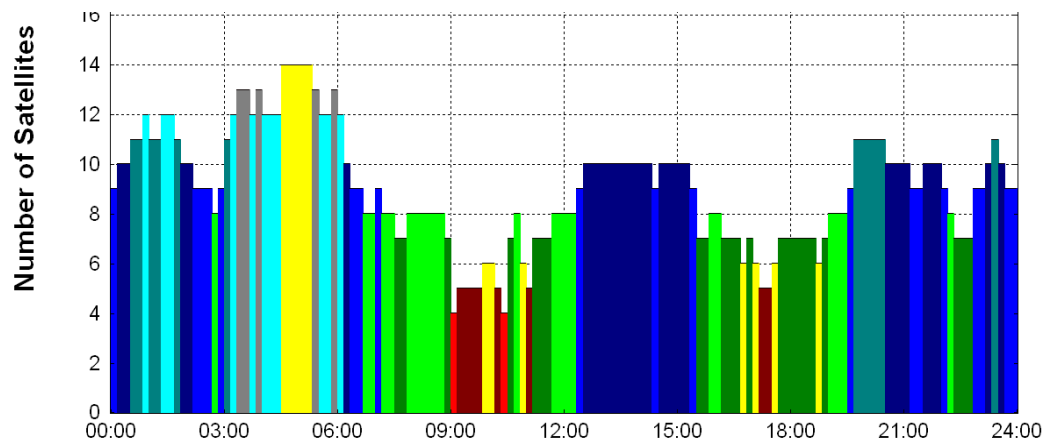
Position: North 45° 46' East 7° 28' Height 700m
Satellites: 35 GPS 28 Glonass 7 [Almanac.alm]

Elevation cutoff 10° Obstacles 0%

Time 23/02/2007 00:00 - 24/02/2007 00:00 (GMT+2.0h)

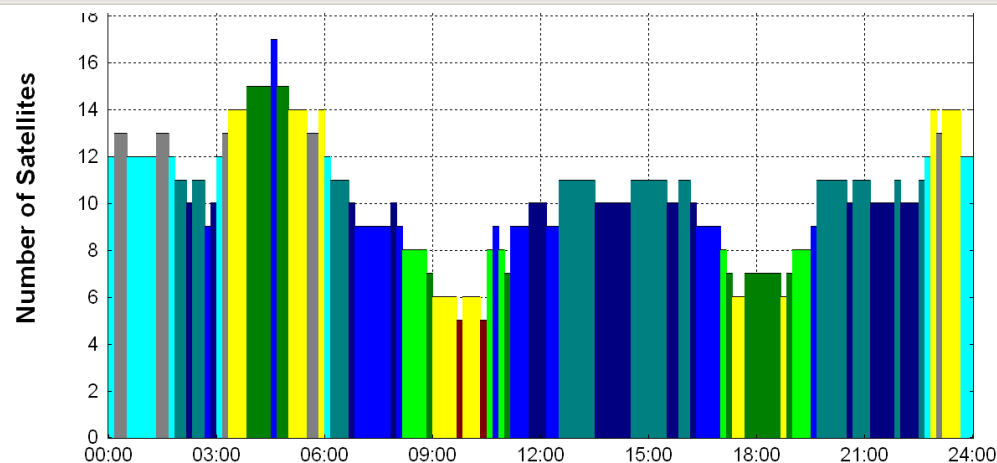


Visibility plot – impact of the obstacles



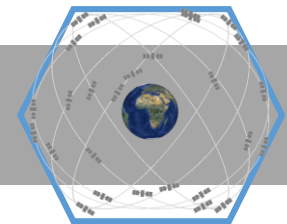
Obstruction

tation North 45° 48' East 7° 28' Height 700m
 atellites 35 GPS 28 Glonass 7 [Almanac.alm]
 Elevation cutoff 10° Obstacles 7%
 Time 23/02/2007 00:00 - 24/02/2007 00:00 (GMT+2.0h)

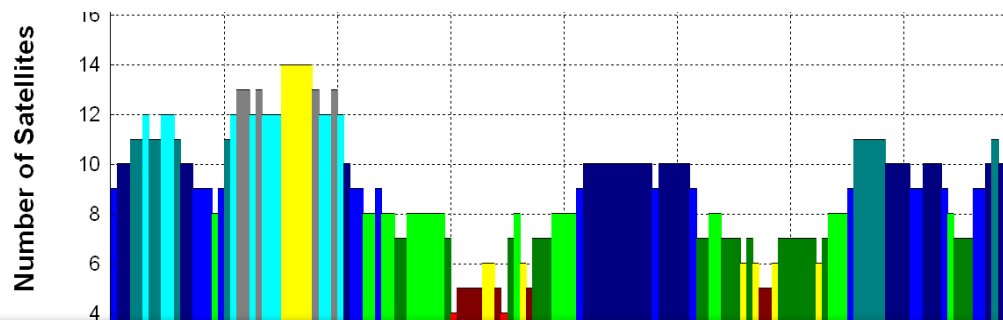


NO obstruction





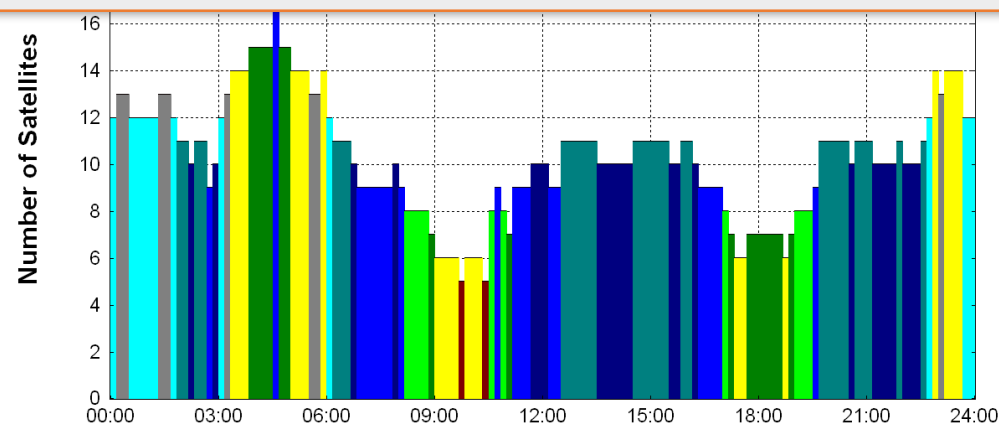
Visibility plot – impact of the obstacles



Obstruction

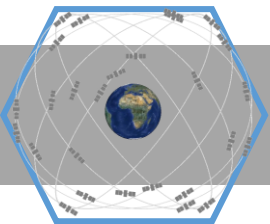
SURVEY:

Is it better to have a 100 m high obstacle 100 m away or a 6 m obstacle 5 m away??

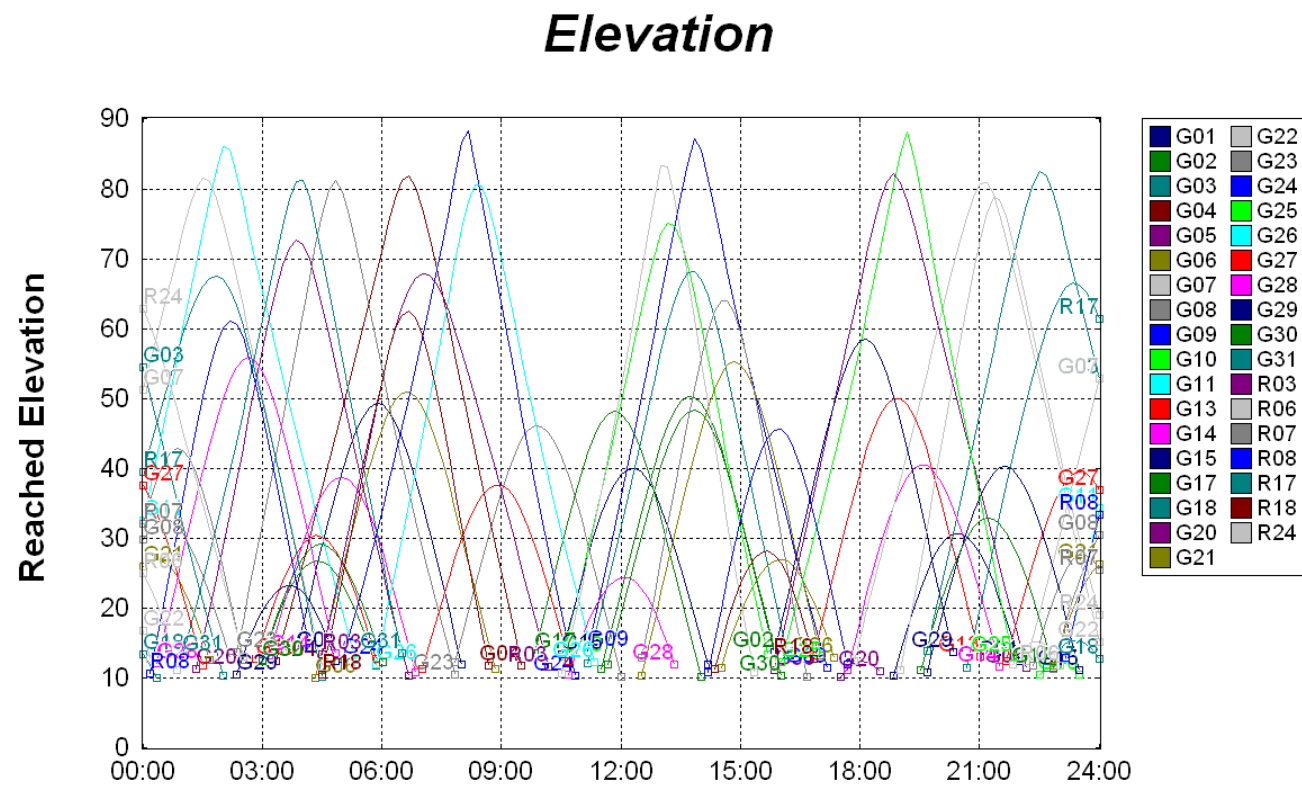


NO obstruction





Elevation plot



Station North 45° 46' East 7° 28' Height 700m
Satellites 35 GPS 28 Glonass 7 [Almanac.alm]

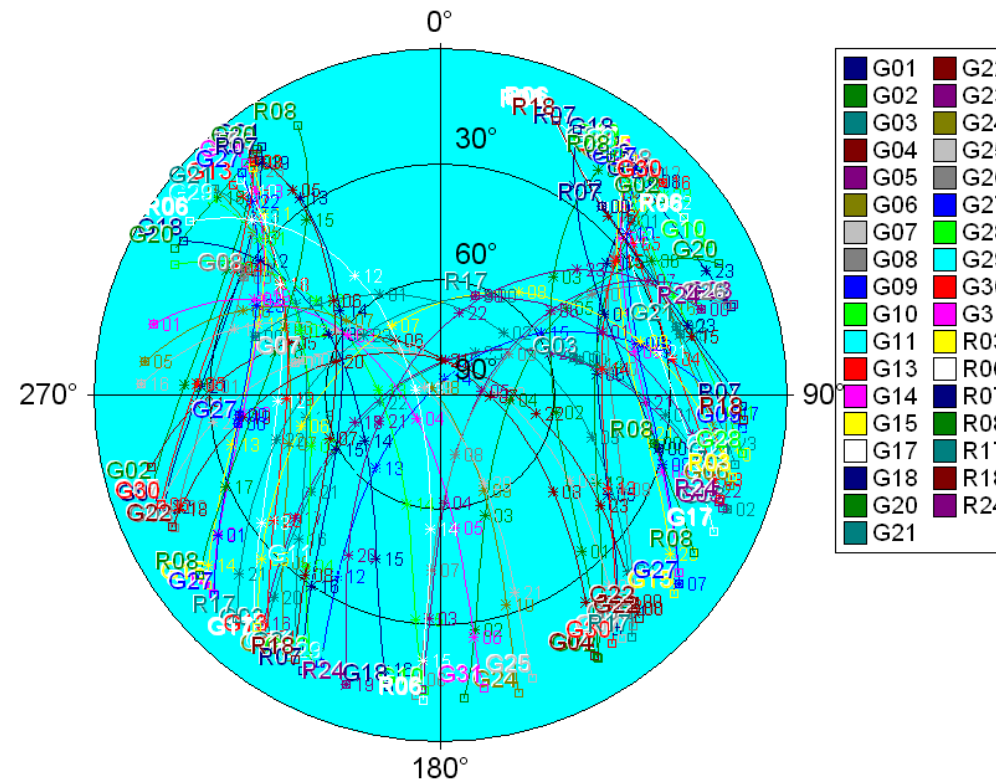
Elevation cutoff 10° Obstacles 0%

Time 23/02/2007 00:00 - 24/02/2007 00:00 (GMT+2.0h)



Skyplot

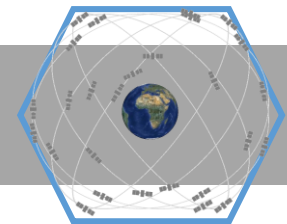
Sky Plot



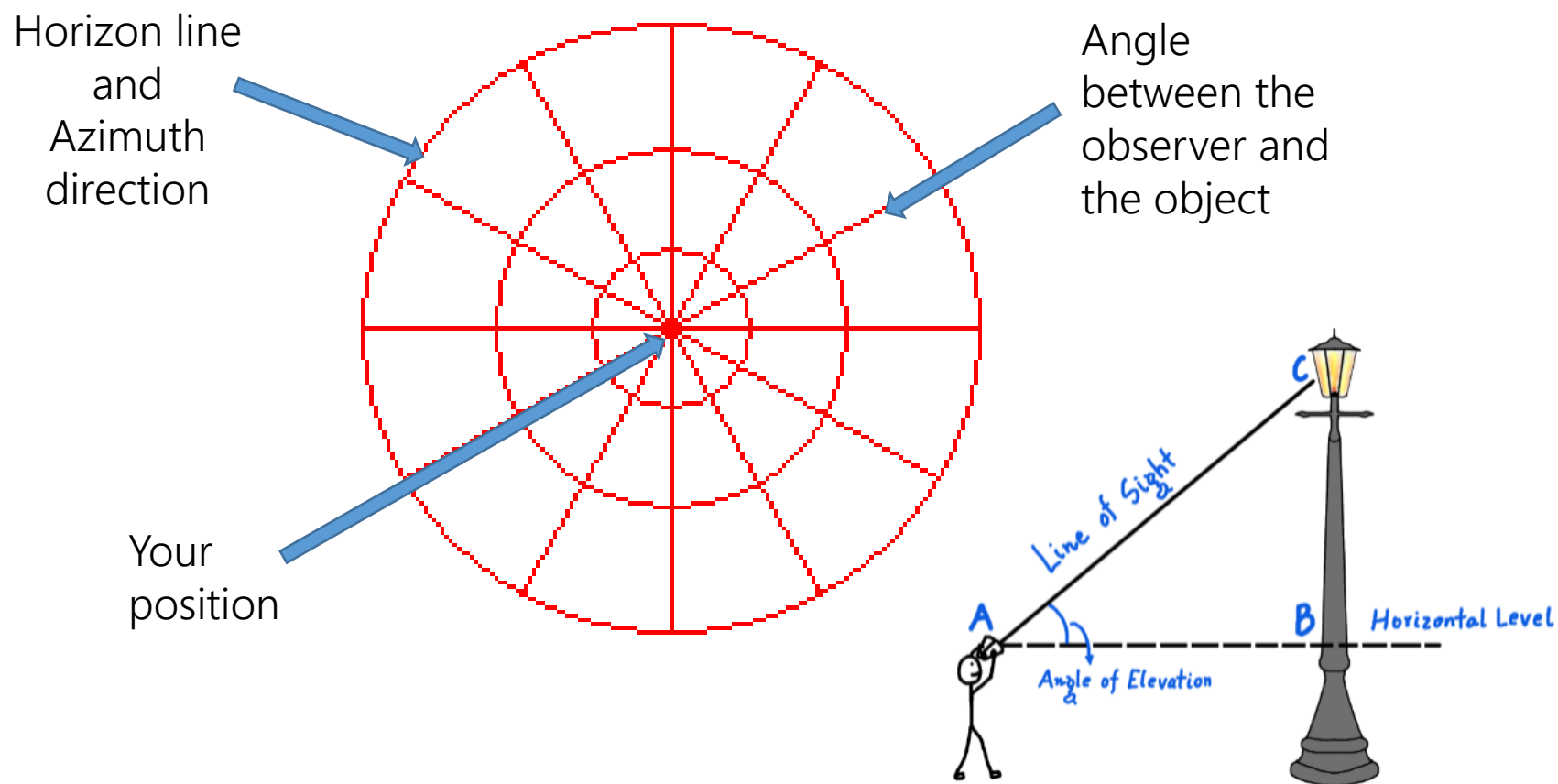
Station North 45° 46' East 7° 28' Height 700m
Satellites 35 GPS 28 Glonass 7 [Almanac.alm]

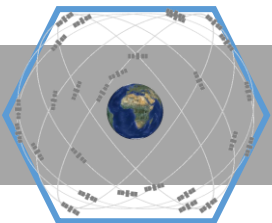
Elevation cutoff 10° Obstacles 0%

Time 23/02/2007 00:00 - 24/02/2007 00:00 (GMT+2.0h)

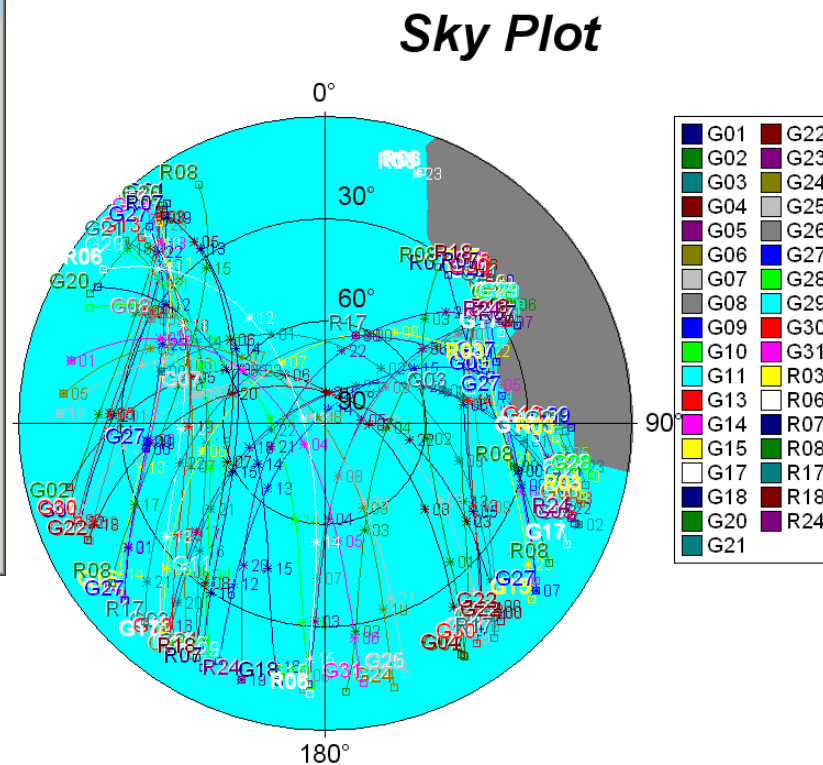
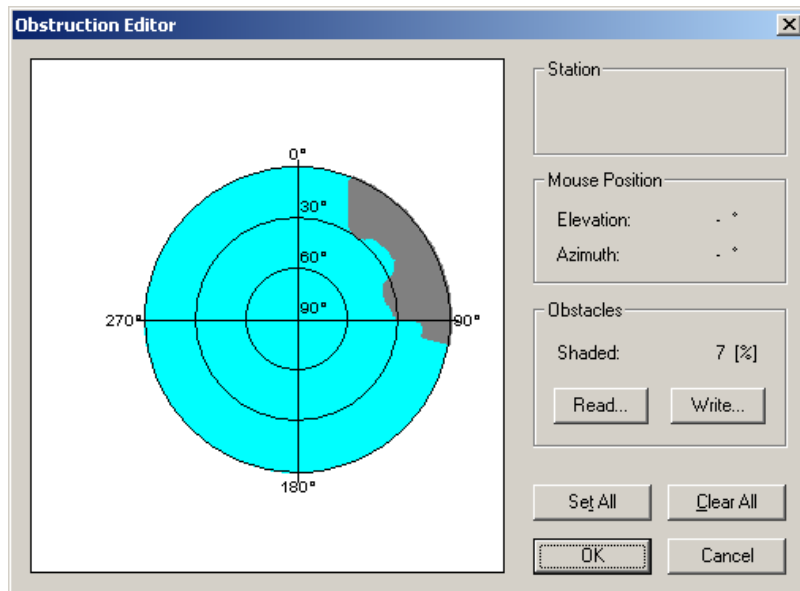


How to read the skyplot?





Skyplot and obstacles

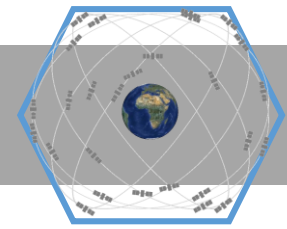


Station North 45° 46' East 7° 28' Height 700m
Satellites 35 GPS 28 Glonass 7 [Almanac.alm]

Elevation cutoff 10° Obstacles 7%

Time 23/02/2007 00:00 - 24/02/2007 00:00 (GMT+2.0h)





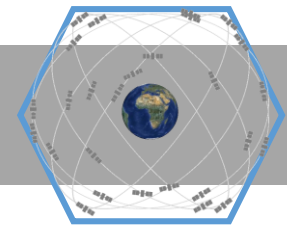
Dilution of precision DOP

$$GDOP = \sqrt{\sigma_T^2 + \sigma_X^2 + \sigma_Y^2 + \sigma_Z^2}$$

$$PDOP = \sqrt{\sigma_X^2 + \sigma_Y^2 + \sigma_Z^2}$$

$$HDOP = \sqrt{\sigma_X^2 + \sigma_Y^2}$$

$$VDOP = \sqrt{\sigma_Z^2}$$



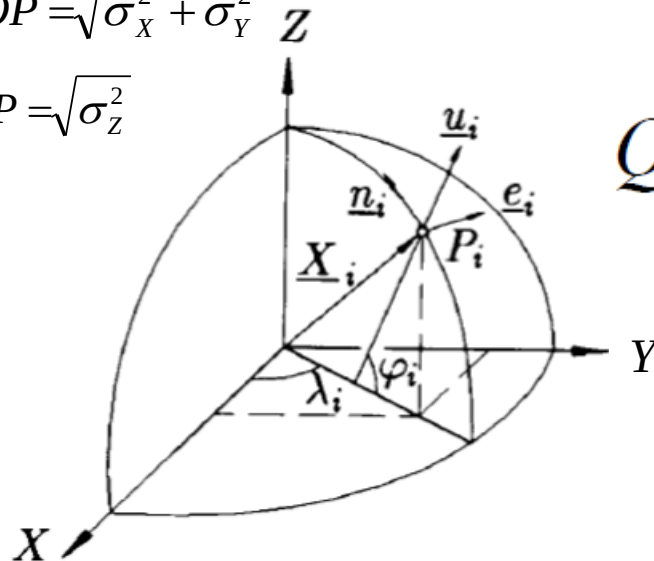
Dilution of precision DOP

$$GDOP = \sqrt{\sigma_T^2 + \sigma_X^2 + \sigma_Y^2 + \sigma_Z^2}$$

$$PDOP = \sqrt{\sigma_X^2 + \sigma_Y^2 + \sigma_Z^2}$$

$$HDOP = \sqrt{\sigma_X^2 + \sigma_Y^2}$$

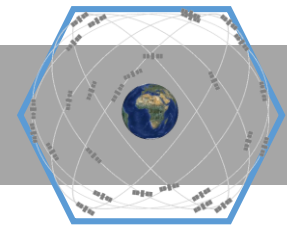
$$VDOP = \sqrt{\sigma_Z^2}$$



Co-factor matrix of site coordinates in a local system

$$Q_{UU} = \begin{bmatrix} \sigma_e^2 & \sigma_{en}^2 & \sigma_{eu}^2 & \sigma_{et}^2 \\ \sigma_{en}^2 & \sigma_n^2 & \sigma_{nu}^2 & \sigma_{nt}^2 \\ \sigma_{eu}^2 & \sigma_{nu}^2 & \sigma_u^2 & \sigma_{ut}^2 \\ \sigma_{et}^2 & \sigma_{nt}^2 & \sigma_{ut}^2 & \sigma_t^2 \end{bmatrix}$$





Dilution of precision DOP

$$GDOP = \sqrt{\sigma_T^2 + \sigma_X^2 + \sigma_Y^2 + \sigma_Z^2}$$

$$PDOP = \sqrt{\sigma_X^2 + \sigma_Y^2 + \sigma_Z^2}$$

$$HDOP = \sqrt{\sigma_X^2 + \sigma_Y^2}$$

$$VDOP = \sqrt{\sigma_Z^2}$$

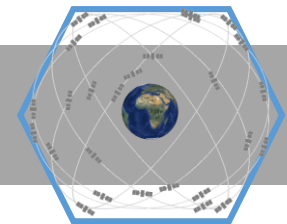
The DOP can be estimated during the planning phase, as the co-factor matrix can be obtained a priori.

$$Q_{xx} = (D^T D)^{-1}$$

$$Q_{UU} = R Q_{XX} R^T$$

$$D = \begin{bmatrix} \frac{X_i^{(0)} - X^{(1)}}{\rho_i^{1(0)}} & \frac{Y_i^{(0)} - Y^{(1)}}{\rho_i^{1(0)}} & \frac{Z_i^{(0)} - Z^{(1)}}{\rho_i^{1(0)}} & -1 \\ \frac{X_i^{(0)} - X^{(2)}}{\rho_i^{2(0)}} & \frac{Y_i^{(0)} - Y^{(2)}}{\rho_i^{2(0)}} & \frac{Z_i^{(0)} - Z^{(2)}}{\rho_i^{2(0)}} & -1 \\ \frac{X_i^{(0)} - X^{(3)}}{\rho_i^{3(0)}} & \frac{Y_i^{(0)} - Y^{(3)}}{\rho_i^{3(0)}} & \frac{Z_i^{(0)} - Z^{(3)}}{\rho_i^{3(0)}} & -1 \\ \vdots & \vdots & \vdots & \vdots \\ \frac{X_i^{(0)} - X^{(8)}}{\rho_i^{8(0)}} & \frac{Y_i^{(0)} - Y^{(8)}}{\rho_i^{8(0)}} & \frac{Z_i^{(0)} - Z^{(8)}}{\rho_i^{8(0)}} & -1 \end{bmatrix} =$$

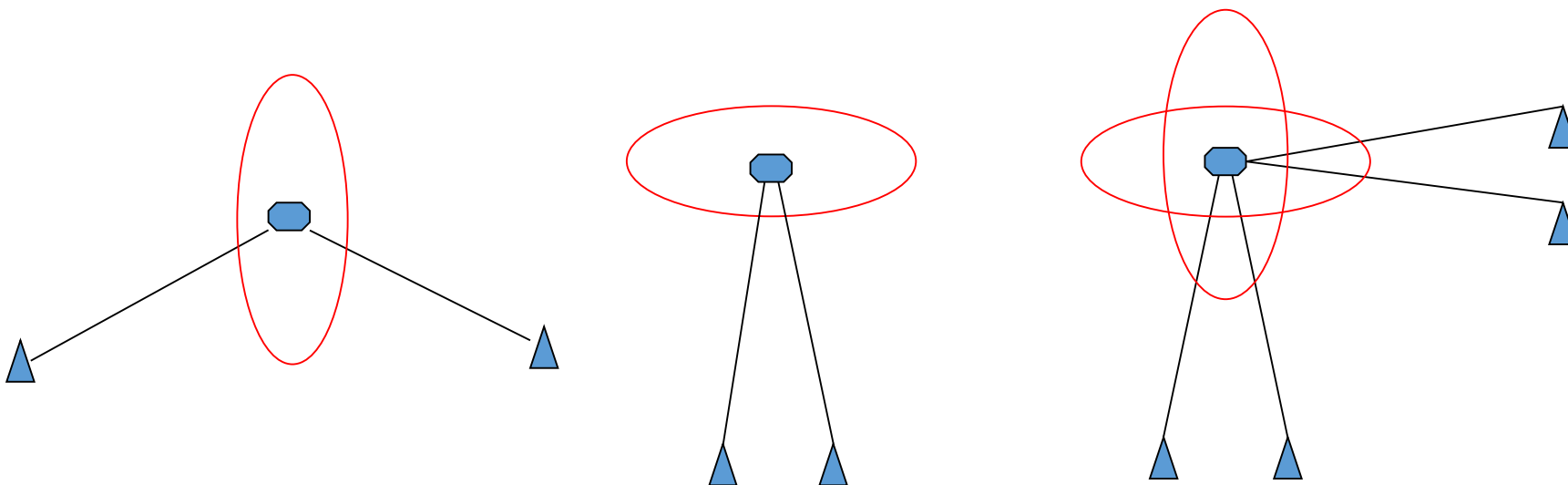


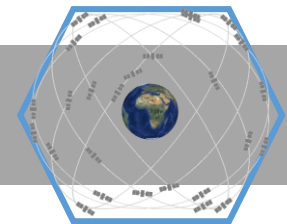


Dilution of precision DOP

Depending on the different locations of the satellites in space, we have a different form of the error ellipse

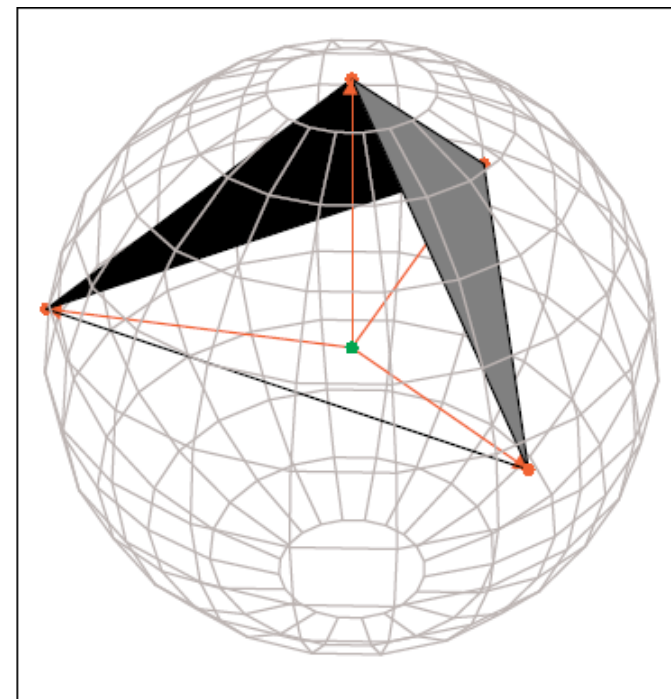
(This ellipse is interpreted analytically from the covariance matrix, in particular by the N^{-1} matrix).

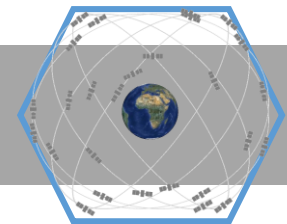




What is the meaning of DOP?

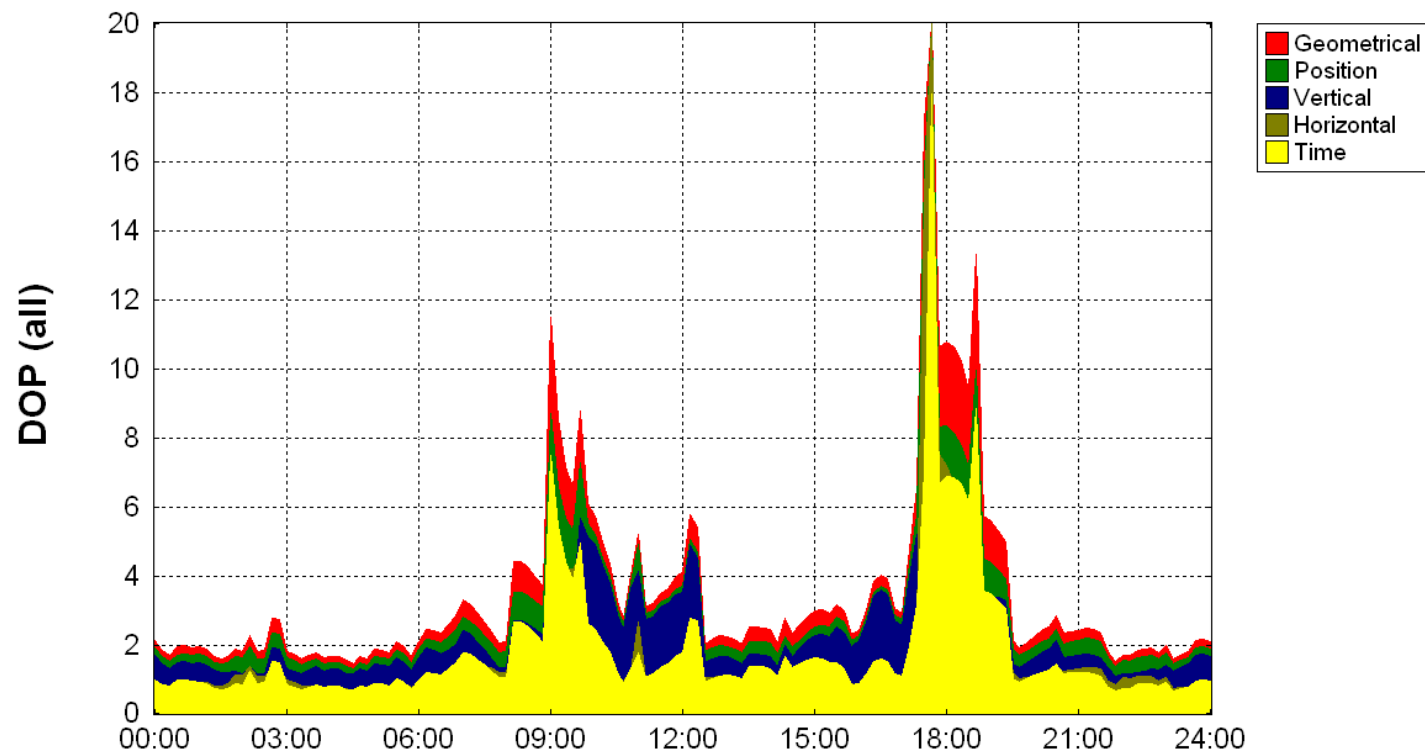
- Considering to have 4 satellites and connect them with the station point. We've a tetrahedron.
- The volume of this tetrahedron is strongly correlated with the value of GDOP, in particular by **increasing the volume decreases the DOP**.
- This means that it is not enough to have a large number of satellites to ensure a good DOP, but also a good satellites disposition.





Dilution of precision DOP

DOP (all)

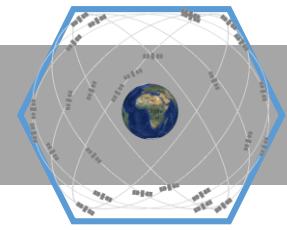


Station North 45° 48' East 7° 28' Height 700m
Satellites 35 GPS 28 Glonass 7 [Almanac.alm]

Elevation cutoff 10° Obstacles 0%

Time 23/02/2007 00:00 - 24/02/2007 00:00 (GMT+2.0h)





How to obtain a good planning

- PDOP → assessments on the geometric configuration
- N° of satellites → duration of the survey
- S/N → quality of the signal
- PDOP and the n° of satellites can be known a priori by good planning and knowledge of any obstructions on the site!!!

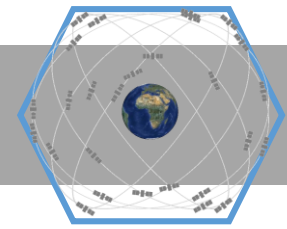
→ An example: TRIMBLE Planning on line

→ <https://www.gnssplanning.com/#/settings>

→ <http://gnssmissionplanning.com/>

See TUTORIAL
PLANNING.pdf



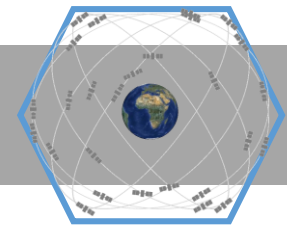


EX 1- Planning and comparison

- a) Realize a planning in four different cities:
 - Student's home town*, Turin, Sydney, Helsinki
- b) Realize the analysis: GPS only, GPS+GLONASS, GNSS (full constellation)
- c) Compare visibility, skyplot, DOP, etc
- d) Using the second tool, include some obstacles and make a new analysis (1 case only) and comparing the results

GOAL: prepare a report discussing the results





EX 2- calculate elevation and azimuth

Input data

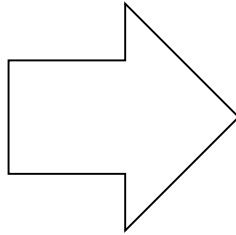
$$X_{\text{sat}} = 17487292,829 \text{ m}$$

$$Y_{\text{sat}} = 6573938,932 \text{ m}$$

$$Z_{\text{sat}} = 17927274,429 \text{ m}$$

$$\text{Latitude (station)} = 44^{\circ} 3' 48.114''$$

$$\text{Longitude (station)} = 7^{\circ} 39' 40.605''$$

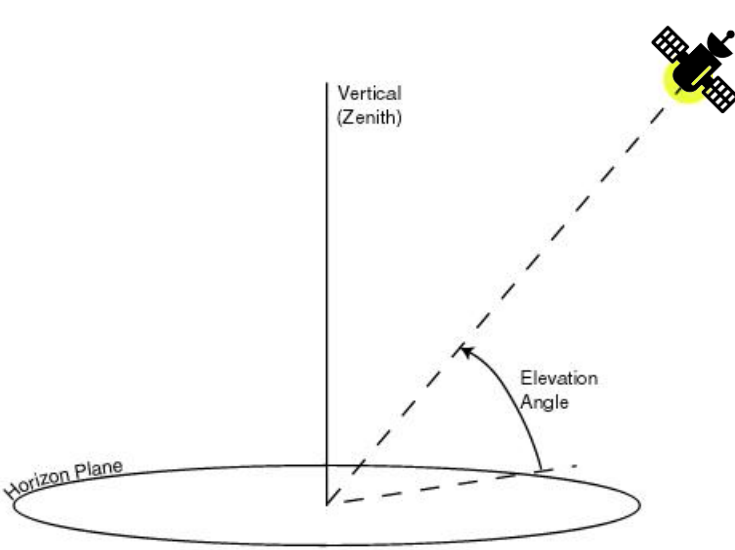


Goal

To define X_i, Y_i, Z_i
station

To estimate AZ, EL of
the SATELLITE

EX 2- calculate elevation and azimuth



$$\begin{pmatrix} e \\ n \\ u \end{pmatrix} = \begin{bmatrix} -\sin \lambda & \cos \lambda & 0 \\ -\sin \varphi \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \\ \cos \varphi \cos \lambda & \cos \varphi \sin \lambda & \sin \varphi \end{bmatrix} \begin{pmatrix} X - X_i \\ Y - Y_i \\ Z - Z_i \end{pmatrix}$$

Where

e, n, u = local coordinates

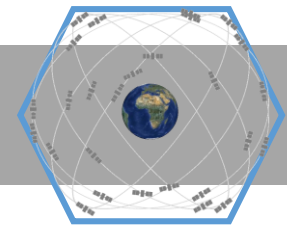
φ, λ = station coordinates (geographic)

X, Y, Z = satellites position (ECEF)

X_i, Y_i, Z_i = station position (ECEF)

$$\text{azimuth} = \text{atn} \frac{u}{\sqrt{n^2 + e^2}}$$

$$\text{elev} = \text{atn} \frac{e}{n}$$



EX 2- annex

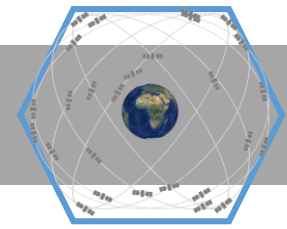
$$\begin{cases} X_i = \frac{a \cdot \cos \varphi \cdot \cos \lambda}{W} \\ Y_i = \frac{a \cdot \cos \varphi \cdot \sin \lambda}{W} \\ Z_i = \frac{a \cdot (1 - e^2) \cdot \sin \varphi}{W} \end{cases} \quad \text{where } W = \sqrt{1 - e^2 \cdot \sin^2 \varphi}$$

WGS84

Parameter	Notation	Value
semi-major axis	a	6378137.0 m
flattening	f	1 / 298.257223

$$e^2 = 1 - b^2/a^2 = 2f - f^2$$

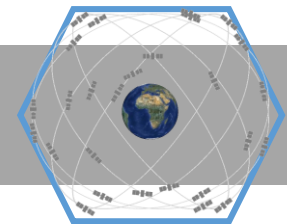




EX 2- calculate elevation and azimuth

- a) Convert latitude and longitude of the station from degrees to radians;
- b) Calculate e^2 and W ;
- c) Calculate X_i , Y_i , Z_i ;
- d) Calculate local coordinates e , n , u ;
- e) Estimate Azimuth and Elevation





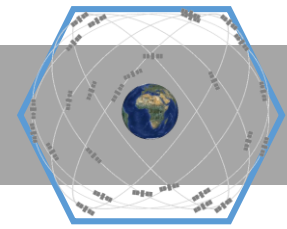
EX 3- DOP estimation

$$Quu = \begin{pmatrix} XDOP^2 & & & \\ & YDOP^2 & & \\ & & VDOP^2 & \\ & & & TDOP^2 \end{pmatrix}$$

Goal

$$\left\{ \begin{array}{l} HDOP = \sqrt{XDOP^2 + YDOP^2} \\ PDOP = \sqrt{XDOP^2 + YDOP^2 + VDOP^2} \\ GDOP = \sqrt{XDOP^2 + YDOP^2 + VDOP^2 + TDOP^2} \end{array} \right.$$





EX 3- DOP estimation

$$\rho_i^j(t) = \sqrt{(X^j(t) - X_i)^2 + (Y^j(t) - Y_i)^2 + (Z^j(t) - Z_i)^2}$$

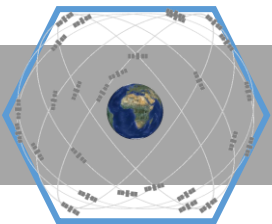
$$D1 = \frac{X_{sat} - X_{rec}}{\rho} \quad D2 = \frac{Y_{sat} - Y_{rec}}{\rho} \quad D3 = \frac{Z_{sat} - Z_{rec}}{\rho} \quad D4 = -1$$

$$D = \begin{bmatrix} D1s1 & D2s1 & D3s1 & -1 \\ D1s2 & D2s2 & D3s2 & -1 \\ \dots & \dots & \dots & \dots \\ D1sn & D2sn & D3sn & -1 \end{bmatrix} \quad \rightarrow \quad Q_{xx} = (D^T D)^{-1}$$

Geographic coordinates
of the station
(expressed in radians)

$$R = \begin{bmatrix} -\sin \lambda & \cos \lambda & 0 \\ -\sin \varphi \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \\ \cos \varphi \cos \lambda & \cos \varphi \sin \lambda & \sin \varphi \end{bmatrix}$$

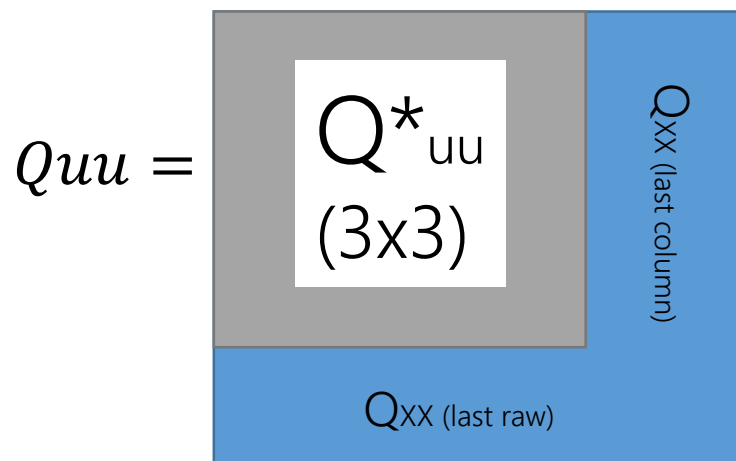




EX 3- DOP estimation

$$Q_{XX}^* = Q_{xx}(1:3; 1:3)$$

$$Q_{UU}^* = R Q_{XX}^* R^T$$



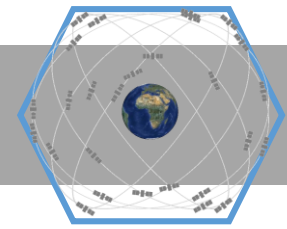
! From the matrix Q_{xx} , a sub-matrix 3×3 is extracted, considering only the first 3 columns and 3 rows.

$Q_{uu} =$

$$\begin{pmatrix} XDOP^2 & & \\ & YDOP^2 & \\ & & VDOP^2 \end{pmatrix} \quad TDOP^2$$

Q_{XX}^*





EX 3- DOP estimation

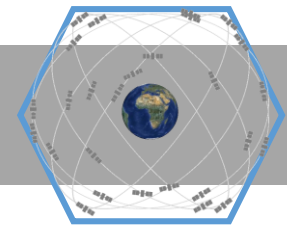
Input data:

Coordinates of the ground receiver $\varphi = 43^{\circ}3'48''$, $\lambda = 7^{\circ}28'41''$, $h = 0m$

Satellite's coordinates (file: pos_sat.txt)

sat	X [m]	Y [m]	Z [m]
1	22504974,806	13900127,123	-2557240,727
2	-3760396,280	-17947593,853	19494169,070
4	9355256,428	-12616043,006	21189549,365
7	23959436,524	5078878,903	-10562274,680
10	10228692,060	-19322124,315	14550804,347
13	23867142,480	-3892848,382	10941892,224
17	21493427,163	-15051899,636	3348924,156
20	14198354,868	13792955,212	17579451,054
23	18493109,722	4172695,812	18776775,463
31	-8106932,299	12484531,565	22195338,169
32	8363810,808	21755378,568	13378858,106





EX 3- DOP estimation

a) Calculate the coordinates X_i , Y_i , Z_i of the receiver;

Tips: If you use Matlab, you can use:

```
wgs84 = wgs84Ellipsoid('meter');
```

```
[x,y,z] = geodetic2ecef(wgs84,fi,lam,h) %mapping toolbox
```

b) Calculate ρ and the D matrix;

c) Calculate R and Q_{xx} matrix;

d) Calculate Q_{uu} matrix and estimate the GDOP, PDOP and HDOP

